

Sean Patten

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PROFESSIONAL SUMMARY

Machine learning researcher and systems engineer with peer-reviewed IEEE publications on ensemble learning and combinatorial fusion, including state-of-the-art sentiment classification research at IBM. Designs and operates production AI-agent infrastructure used daily: a multi-device agent fleet platform with automated review gates, sub-millisecond C tooling, and an unattended trading system running with personal capital. M.S. in Computer Science, Fordham University.

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern, IBM

May 2023 - May 2024

- Led the research and development of a novel ensemble system for sentiment analysis, integrating models using combinatorial fusion.
- Achieved state-of-the-art (SOTA) accuracy of 97.072% on the IMDB dataset, surpassing existing benchmarks.
- Research resulted in a peer-reviewed publication (PICom 2025) and formed the basis of a Master's thesis.
- Collaborators: Dr. Pin-Yu Chen, Dr. D. F. Hsu, Dr. C. Schweikert.

Member, Laboratory for Informatics and Data Mining (Dr. D. F. Hsu), Fordham

2024 - Present

- Fordham CIS research lab for machine learning, data mining, and information fusion, centered on combinatorial fusion analysis.
- Research Team Lead (2026 - Jun 2026): led the research team (Fordham LIDM students with Dr. Samuel Yen-Chi Chen, first author of the original QLSTM paper) building a reproducible quantum-classical Combinatorial Fusion Analysis pipeline: a 40-model pool of 26 quantum-circuit feature maps (including gradient-trained variational quantum classifiers) and 14 classical baselines, with cognitive-diversity-based model selection; work produced the accepted QCE 2026 paper (see Publications).

Software Developer (Contract), Social Science Research Council

October 2024 - January 2025

- Developed a data collection and analysis pipeline to investigate trends in language model adoption and usage on the Hugging Face platform, supporting SSRC research initiatives.
- Work under direction of Tim O'Reilly and Ilan Strauss.

AI Engineer Consultant, Aspire Aviation

January 2025 - Present

- Developed predictive models for aircraft part failure and automated internal document generation systems using Python and LLMs.
- Constraint-based pilot crew scheduler in live operational use.

PUBLICATIONS

Xu, J., Wu, Y., Chen, S. Y.-C., Patten, S., Schweikert, C., & Hsu, D.F. (2026). Diversity-Aware Ensemble of Quantum-Classical Models via Combinatorial Fusion. Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE 2026), Quantum Applications track, Toronto (accepted, not yet published).

Patten, S., Chen, P.-Y., Schweikert, C., & Hsu, D.F. (2025). Enhancing Sentiment Classification with Machine Learning and Combinatorial Fusion. Proceedings of the 23rd IEEE International Conference on Pervasive Intelligence and Computing (PICom 2025). [\[Link\]](#) [\[Code\]](#)

Erdős Number: 2 (via Dr. D. F. Hsu)

PROJECTS

a: agent fleet & automation platform

<https://github.com/seanpattencode/a>

- Multi-device agent and automation platform run as daily infrastructure: compiled-C CLI, systemd-timer job hub, git-synced state across a fleet of machines, orchestrating multiple AI coding agents with automated review gates.
- Every interactive tool held to a 1.0 ms latency budget by a daily watchdog that PTY-measures cold render and per-keystroke repaint and emails on regression; per-tool ok/fail run logs and failure alerts across the fleet.
- C hot paths validated entry-for-entry against the reference Python implementation by a scripted byte-identical diff harness.
- Prompt management as infrastructure: every agent launch composes git-versioned prompt files (system, project, and role layers); prompt edits are reviewed and diffed like code, with a human gate before adoption.
- Agent evaluation beyond accuracy: anonymized multi-model judge panels (question to candidates to vote to mechanical tally) and independent two-model objection checks review outbound artifacts before anything ships.
- Daily-use voice interface: microphone capture to Parakeet (sherpa-onnx) speech recognition to LLM to spoken TTS reply; fleet-wide dictation runs the same stack.

civ: offline full-text search over a civilization-scale corpus

- ~600 GB local corpus (Wikipedia, all 70,000 Project Gutenberg books, 2M arXiv papers) searchable in ~250 ms via a 47 GB SQLite FTS5 index; one-command source acquisition and indexing (C + Python).

SuperCowsay: cowsay rewritten in assembly, at the Linux exec floor

<https://github.com/seanpattencode/SuperCowsay>

- 629-byte hand-written ELF (assembler emits the executable directly, no linker): runs 1.3 microseconds above the kernel's process-spawn floor, measured against an exit-only binary, at 2 syscalls and 145x the original Perl; a second build is byte-identical to the Perl original across 344 differential-fuzz cases (stdout, stderr, exit code) at 35.7x its speed. All rows measured in one pinned hyperfine session.

e: terminal text editor in C, faster than Torvalds's uemacs

<https://github.com/seanpattencode/e>

- Self-compiling single-file C editor (sh e.c builds it): 0.37 ms startup, 2.8x faster than Torvalds's uemacs of the same micro-emacs heritage; opens a 5,600-line file in 1.2 ms (nvim: 223 ms) and renders an interactive directory browser 3x faster than ls prints filenames (hyperfine-measured).

seanpatten.com: performance-minimal personal website

<https://github.com/seanpattencode/seanpattencode.github.io>

- Single ~5 KB file scoring 100/100 on PageSpeed: no images, external resources, or favicon request; native system fonts; a strict subset of the original 1991 HTML tag set, rendering correctly in everything from Lynx and 1993 Mosaic to current browsers.
- Hosts live two-player Pong, playable over the internet with no game server: WebRTC pairs players peer-to-peer at near-frame latency. Play at <https://seanpatten.com/games/pong/>
- Serves a real AI chat service: Gemma 4 running on a home machine (CPU only), replies streamed live to the browser. Chat at <https://seanpatten.com/chat/>
- Self-hosted dynamic pages with server state, served from a home machine through a Cloudflare tunnel by the a platform above.

Custom automated trading system

- Automated market scanning, armed alert watches, pre-queued order workflows, and an append-only audit log of every action (C + Python), running unattended across a self-managed multi-device fleet; in production with personal capital.
- Strategy proprietary; engineering details available in conversation.

RESEARCH OBJECTIVE

Training, combining, and architecting ensemble strategies to improve performance, safety, and cost effectiveness.

EDUCATION

M.S. in Computer Science

August 2022 - May 2024

Fordham University Graduate School of Arts and Sciences, Lincoln Center, New York

- GPA: 3.667/4.0
- **Thesis:** Enhancing Sentiment Classification with Machine Learning and Combinatorial Fusion
- **Advisor:** Dr. D. F. Hsu

Relevant Coursework: Artificial Intelligence, Deep Learning, NLP, Machine Learning, Data and Information Fusion (Ensemble Learning)

B.S. in Computer Science & B.S. in Business Administration

August 2018 - May 2022

Fordham University, Gabelli School of Business, Bronx, New York

Relevant Coursework: Core Computer Science, Core Business Classes, Concentration in Entrepreneurship

AWARDS & HONORS

- National Merit Scholarship Semifinalist
- Johns Hopkins Center for Talented Youth Student
- We the People Competition 2018 New York State Winner & Nationals Finalist

TECHNICAL SKILLS

AI Agents: Claude Code, OpenAI Codex, Gemini CLI, Grok, Google Antigravity; used together in multi-agent workflows: worktree orchestration, review gates, MCP, browser automation, multi-device agent fleets, LLM fine-tuning, RAG, voice pipelines (STT/TTS)

Languages & Frameworks: Python, C, C++, PyTorch, Tensorflow, scikit-learn, Java, Javascript, SQL, CUDA, Assembly

Platforms & Tools: Huggingface, Git, WebRTC, ffmpeg, sherpa-onnx (speech recognition), Qiskit

TEACHING

Club President, Fordham Digital Business Society

2021-2022

Helped Fordham students learn practical digital skills. Held events on learning Python, personally teaching the basics to beginners, and increased attendance by over 1,000% compared to previous years.

QUANTUM COMPUTING

Quantum Computing Researcher, Fujitsu Quantum Simulator Challenge

January 2026 - Present

- First author of the Asia University team's accepted proposal (Q-Compress: Variational Entanglement Profiling for Extreme LLM Compression); currently running LLM weight-compression experiments on Fujitsu's 40-qubit tensor network simulator, a 1,024-node FX700 (A64FX) supercomputer cluster, via the QARP SDK and Qulacs.
- Maps transformer weight super-blocks onto qubit registers and uses entanglement profiling (Von Neumann entropy across bipartitions) to derive non-uniform, entropy-aware tensor-network compression topologies, targeting 80%+ memory reduction with under 2% accuracy loss against SVD-LLM, ASVD, SparseGPT, and CompactifAI baselines.